

PULSED EMITTER – 20 kHz / 5 us / ~500 mA

Emitter loop is local: C20 supplies the 5 us / 500 mA pulse so the cable carries only the ~50 mA average. R19 damps the gate edge; R20 holds Q1 off while the main board is unpowered or the cable is unplugged.

TRANSIMPEDANCE + x12 AC GAIN

The circuit diagram illustrates a transimpedance amplifier with a gain of 12. It is powered by a +3V3 supply. The input signal is provided by a photodiode (PD1, BPW34S) connected to the inverting input (PD_TIA_IN) of the first operational amplifier (U8A, TLV9062IDR). The non-inverting input (OPT_VRE6) of U8A is grounded. The feedback network of U8A consists of a capacitor (C21, 10pF) and a resistor (R21, 10k) connected in parallel between the output (TIA_OUT) and the inverting input. The output of U8A (TIA_OUT) is connected to the non-inverting input (AMP_INV) of the second operational amplifier (U8B, TLV9062IDR). The inverting input (AMP_INV) of U8B is connected to ground through a resistor (R22, 10k). The feedback network of U8B consists of a capacitor (C22, 10nF) and a resistor (R23, 120k) connected in parallel between the output (OPT_AMP) and the inverting input. The final output of the circuit is OPT_AMP.

The diagram illustrates a threshold comparator circuit with hysteresis, implemented using a TS3021NLT op-amp (U4). The circuit is powered by a +3V3 supply and includes a 100nF decoupling capacitor (C25).

Input Stage: The input signal, labeled **COMP_THRESHOLD**, is connected to the non-inverting input (pin 3) of the op-amp through a 22k resistor (R26). A 10k resistor (R27) connects the non-inverting input to ground.

Feedback and Output Stage: The output of the op-amp (pin 1) is labeled **COMP_OUT**. This output is connected to the inverting input (pin 2) through a 470k resistor (R28), creating a positive feedback loop (Schmitt trigger). Additionally, the output is connected to ground through a 33R resistor (R32), labeled **OPT_COMP**.

Power and Grounding: The op-amp is powered by a +3V3 supply (pin 5) and has its ground connection (pin 4, labeled **OPT_AMP4**) connected to ground. A 100nF capacitor (C25) is connected between the +3V3 supply and ground to filter noise.

1.65 V REFERENCE

CABLE TO MAIN BOARD

J5
TO MAIN BOARD

7 +3V3
6 ORF_COMP
5 ORF_LED_EN
4 ORF_LED_EN
3 ORF_LED_EN
2 SYS_SW
1 SYS_SW

+3V3
SYS_SW
GND